

MetaRadar — 4-Week Prototype Development Timeline

Technical Implementation Sequence for Near-Real-Time Competitive Intelligence Prototype

Target Domain: Haemophilia / Rare Disease
Architecture: Next.js 16 + FastAPI + PostgreSQL



Week 1 — Foundation

Focus Areas:

Architecture, data model, API contracts, configuration, dev environment.

Key Deliverables:

- **Entity Schema:** Database models for signals, assets, sources, and 384-dim vector store.
- **API Contracts:** OpenAPI schema and unified TypeScript contract generator.
- **Domain Config:** Config loader ('haemophilia.yaml') for assets, functions, and rules.
- **Dev Setup:** Baseline workspace configuration, Dockerfiles, and test suite.

Week 2 — Ingestion

Focus Areas:

Source connectors, query configuration, incremental fetching, Bronze persistence, deduplication.

Key Deliverables:

- **Source Connectors:** Async adapters for PubMed, ClinicalTrials.gov, NewsAPI, FDA, and EMA.
- **Bronze Layer:** Verbatim raw payload persistence with SHA-256 content hashing.
- **Deduplication:** Stable fingerprinting (PMID, NCT ID, Reg ID) and chunking.
- **Syndication Logic:** Polling controls, rate limits, and source freshness reporting.

Week 3 — Intelligence

Focus Areas:

Normalization, evidence processing, signal pipeline, provider integration, privacy/safety boundaries.

Key Deliverables:

- **Entity Normalization:** Entity extraction and domain ontology mapping.
- **Red-Team Engine:** Pairwise trial contradiction evaluator across 19 safety/efficacy rules.
- **Stateful Workflow:** 10-node pipeline orchestrating evidence confluences and lifecycles.
- **Provider Matrix:** Local Gemma 3 4B, Grok privacy gate, and BART fallback.

Week 4 — Integration

Focus Areas:

Frontend/API integration, testing, observability, runtime verification, bug fixing, prototype demo.

Key Deliverables:

- **UI Integration:** Next.js App Router UI connected to backend '/api/v1' REST services.
- **Four-Question UX:** Interactive decision panels and Ask Athena prompt interface.
- **Verification:** End-to-end evidence story testing (mim8 / Hemgenix scenario).
- **Demo Readiness:** Quality gate verification, runtime testing, and prototype presentation.